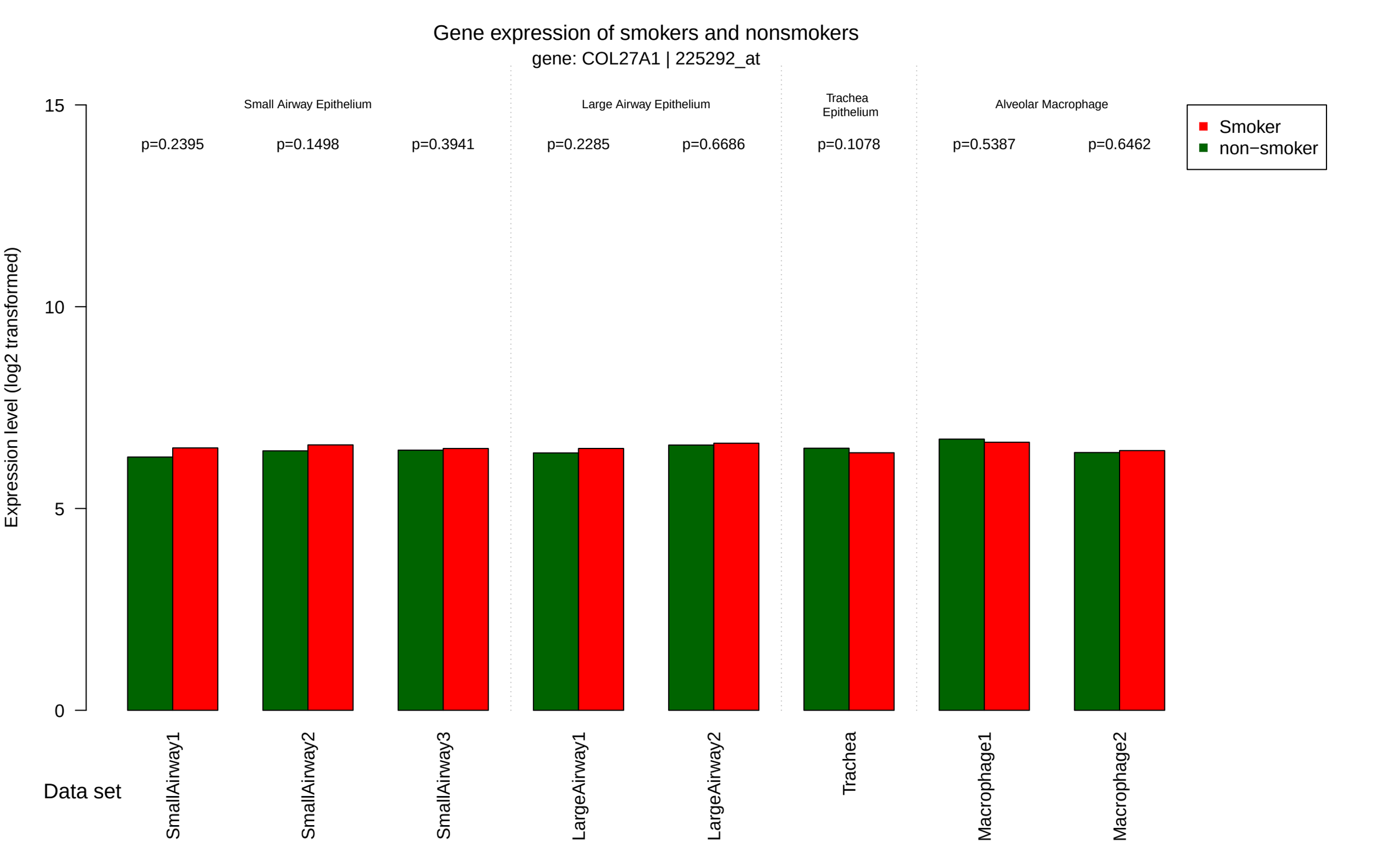
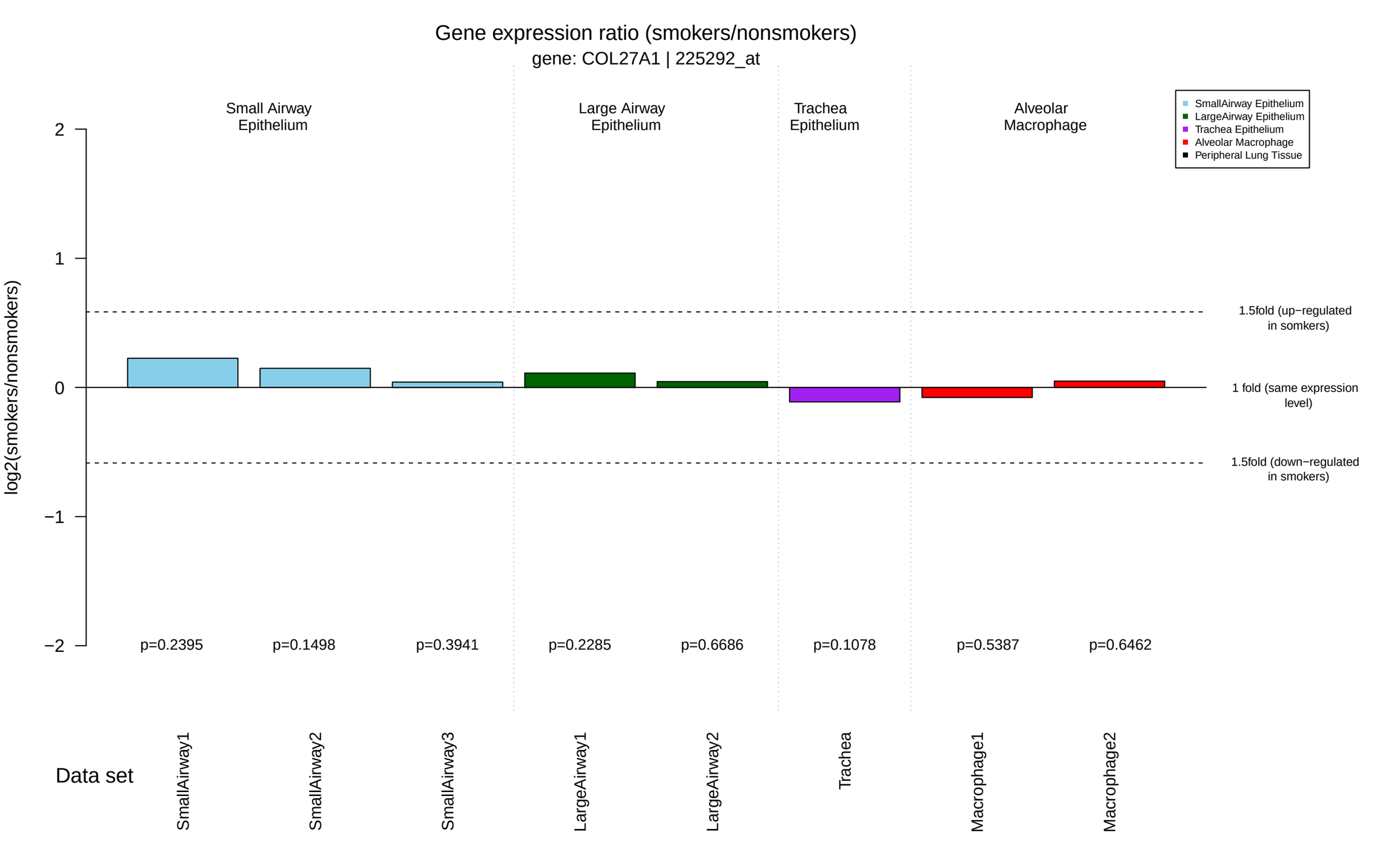
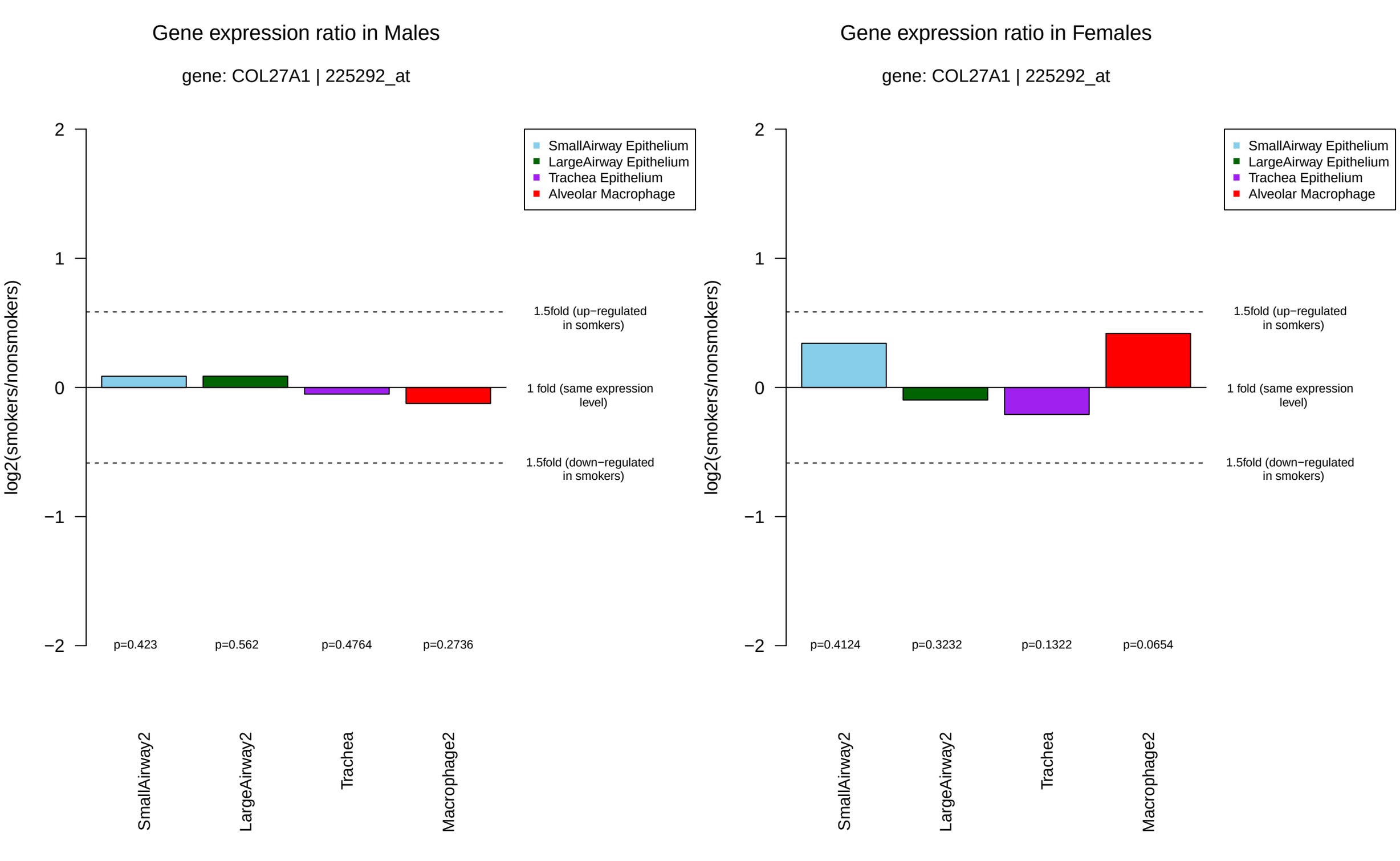
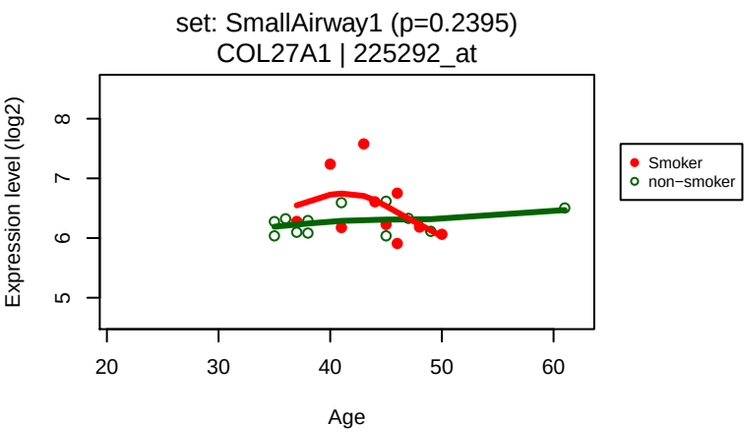
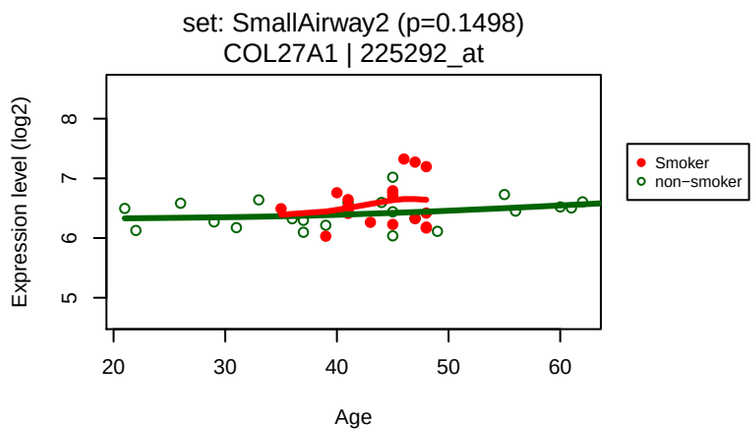
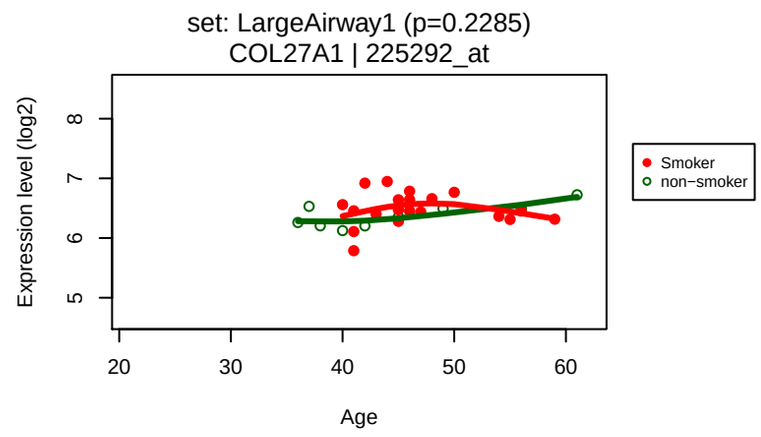

Gene expression of smokers and nonsmokers
gene: COL27A1 | 225292_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2395
p=0.1498
p=0.3941
p=0.2285
p=0.6686
p=0.1078
p=0.5387
p=0.6462
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

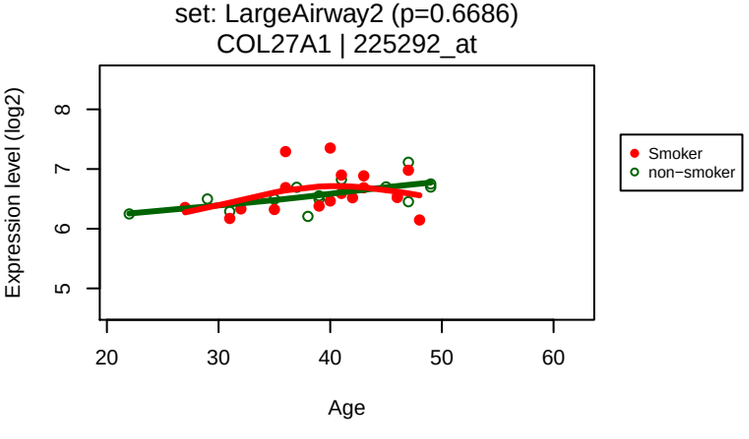

Gene expression ratio (smokers/nonsmokers)
gene: COL27A1 | 225292_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2395
p=0.1498
p=0.3941
p=0.2285
p=0.6686
p=0.1078
p=0.5387
p=0.6462
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

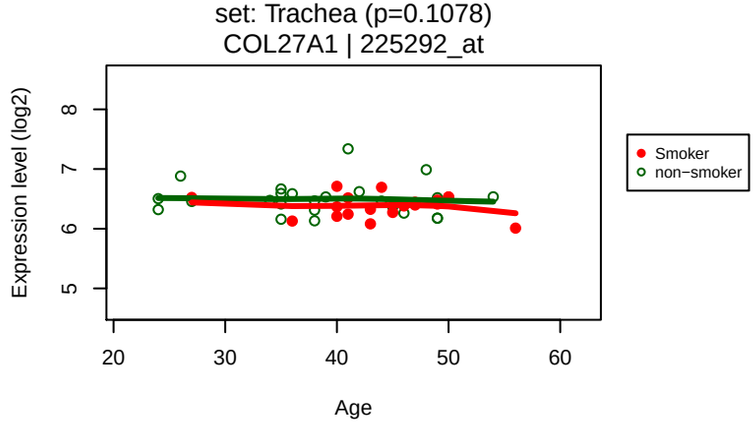

Gene expression ratio in Males
gene: COL27A1 | 225292_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.423
p=0.562
p=0.4764
p=0.2736
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL27A1 | 225292_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4124
p=0.3232
p=0.1322
p=0.0654
SmallAirway2
LargeAirway2
Trachea
Macrophage2

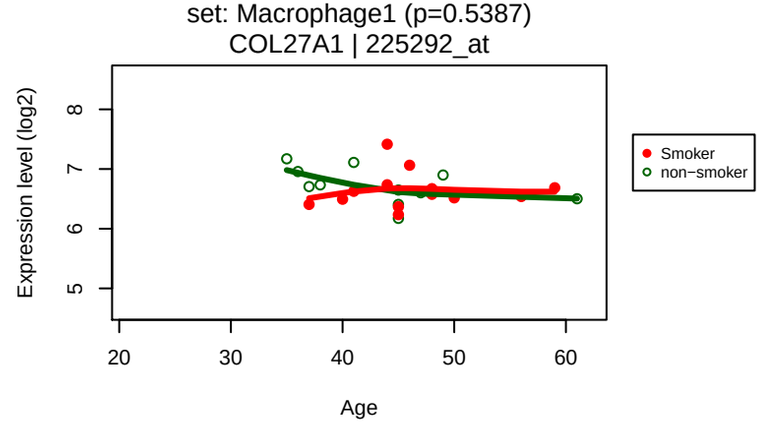

set: SmallAirway1 (p=0.2395)
COL27A1 | 225292_at
Expression level (log2)
Age
Smoker
non−smoker

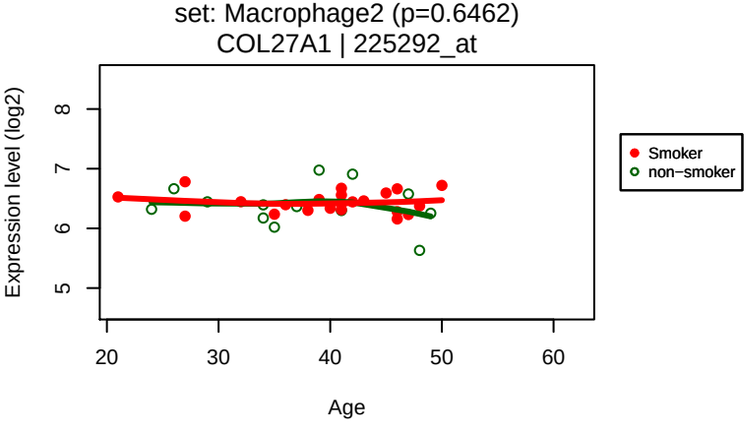

set: SmallAirway2 (p=0.1498)
COL27A1 | 225292_at
Expression level (log2)
Age
Smoker
non−smoker

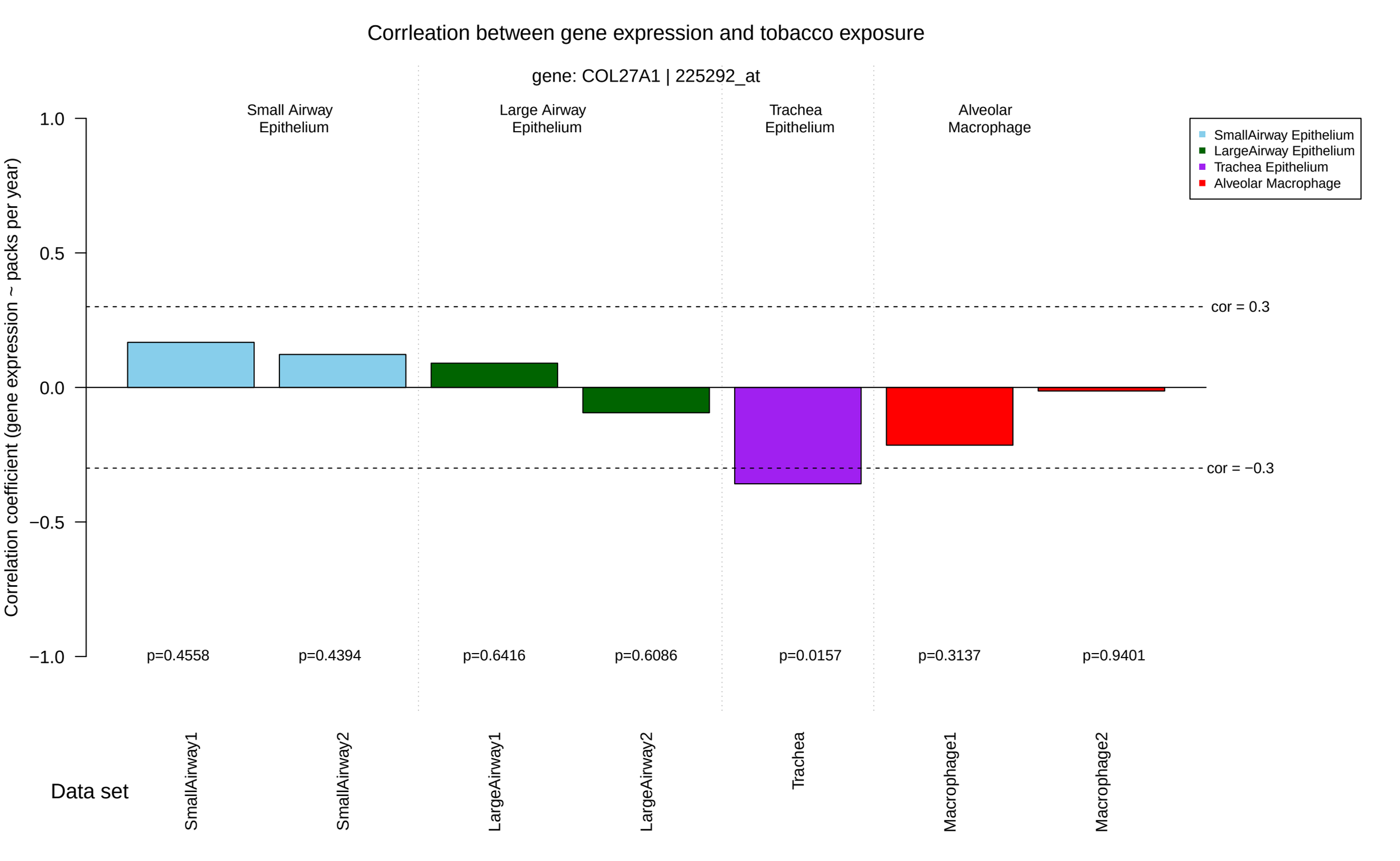

set: LargeAirway1 (p=0.2285)
COL27A1 | 225292_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6686)
COL27A1 | 225292_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1078)
COL27A1 | 225292_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5387)
COL27A1 | 225292_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6462)
COL27A1 | 225292_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL27A1 | 225292_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4558
p=0.4394
p=0.6416
p=0.6086
p=0.0157
p=0.3137
p=0.9401
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2